Lab – Deployment Slots (Azure App Service)

Summary

This lab demonstrates how to use **Deployment Slots** in **Azure App Service** to deploy a new version of a web application without affecting the production environment. You will create a staging slot, deploy the updated application to it, verify that it works correctly, and then swap it with the production slot for a seamless deployment with minimal downtime.

Understanding

Deployment Slots in Azure App Service allow you to deploy and test new versions of your application in a separate environment before making them live. Each slot is a live App Service with its own hostname. Once testing is complete, you can perform a **Swap** operation, which exchanges the content and configuration (except slot-specific settings) between the staging slot and the production slot. This enables near zero-downtime deployments and provides an easy rollback option if any issues occur after deployment.

Example:

Suppose your company has an online shopping website running in the **Production** slot. A new version of the application has been developed. Instead of deploying directly to Production, you create a **Staging** slot and deploy the new version there. After testing the application using the staging URL, you swap the Staging slot with Production. Users immediately see the new version without experiencing downtime. If any problem occurs, you can swap back to restore the previous version.

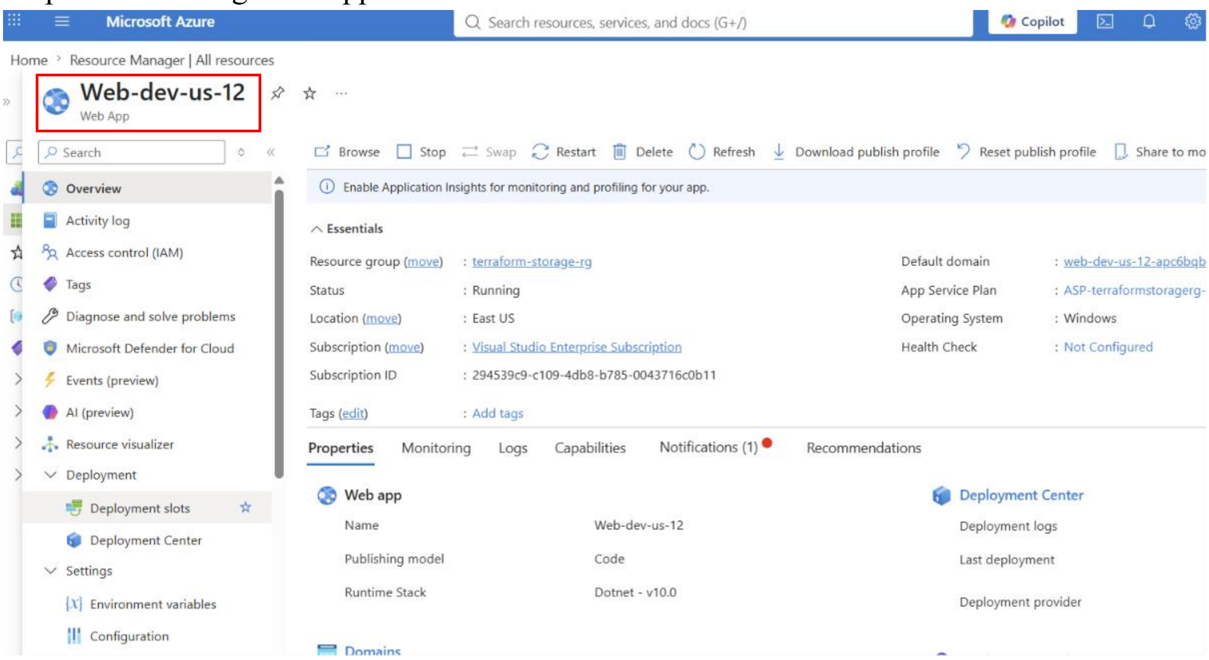
Lab Steps

Step 1: Open Azure Portal

• Sign in to the Azure Portal.

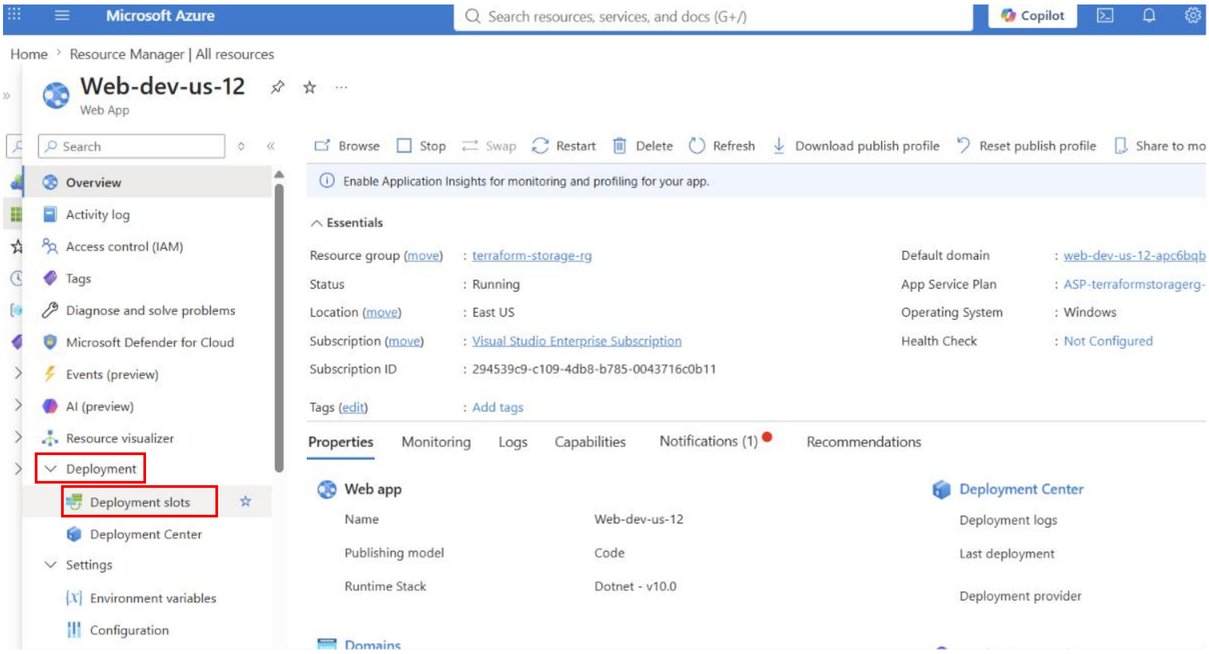
• Navigate to **App Services**.

• Open the existing Web App.



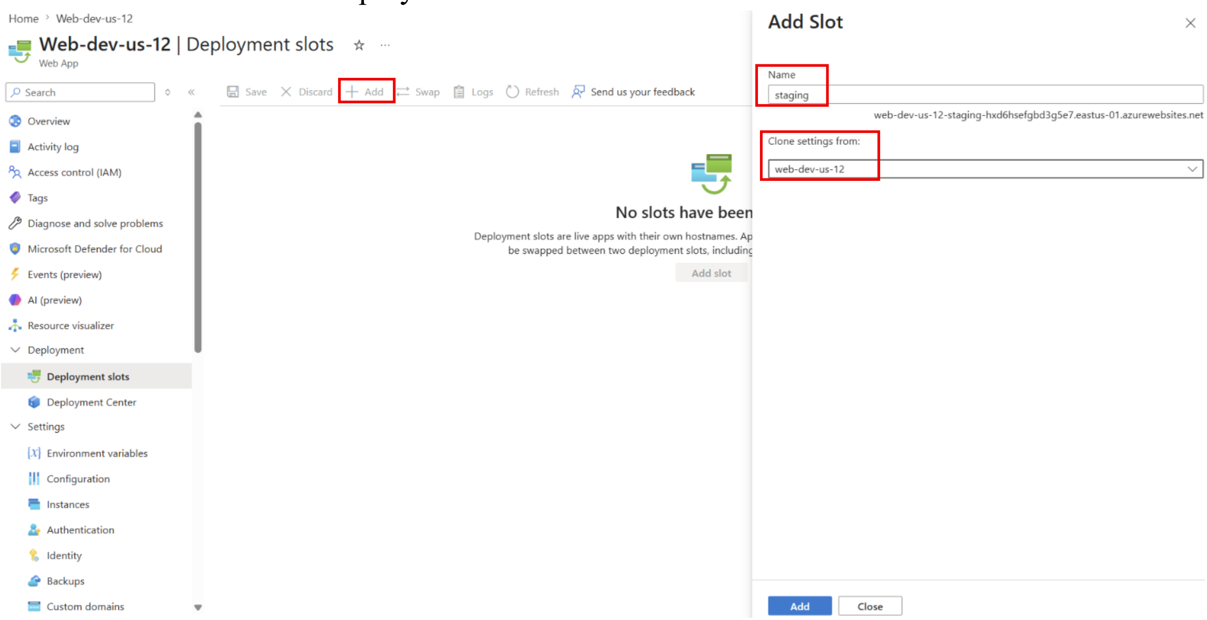
Step 2: Open Deployment Slots

- Navigate to **Deployment**.
- In the left navigation pane, select **Deployment slots**.
- View the existing Production slot.



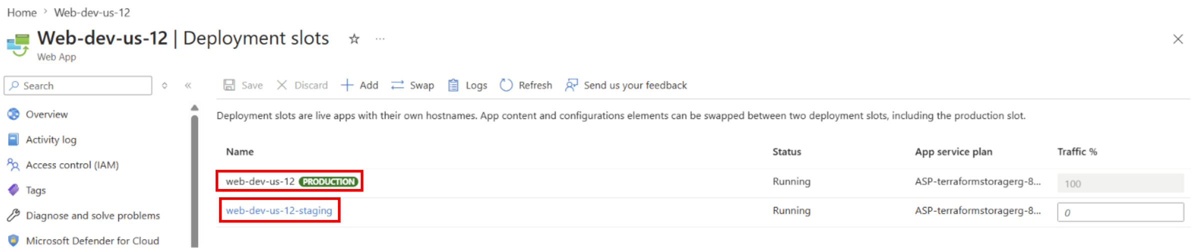
Step 3: Add a New Slot

- Click **Add Slot**.
- Enter the slot name (for example, **staging**).
- Choose whether to clone settings from the **Production slot.**
- Click **Add** to create the deployment slot.



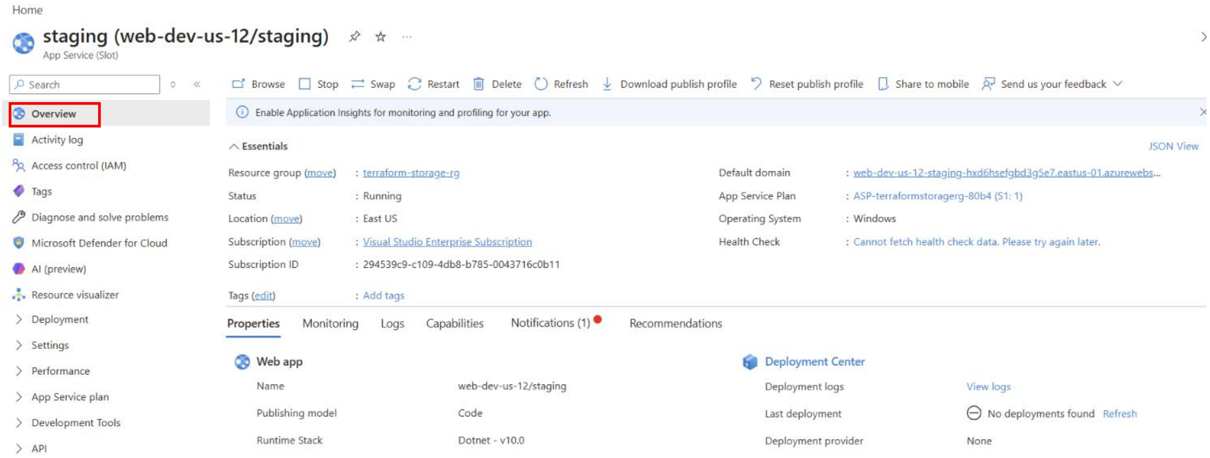
Step 4: Wait for Slot Creation

- Azure creates the new deployment slot.
- Verify that the staging slot appears in the Deployment Slots list.



Step 5: Open the Staging Slot

- Click the newly created **Staging** slot.
- Review its Overview page.

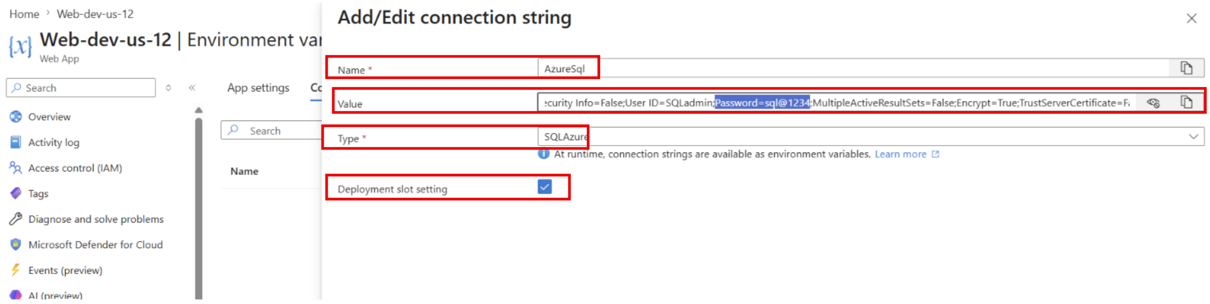


Step 6: Add connecting string

- Go to the **Azure Own SQL Database.**
- Navigate to **Connecting string.**
- Copy **ADO.NET (SQL authentication).**

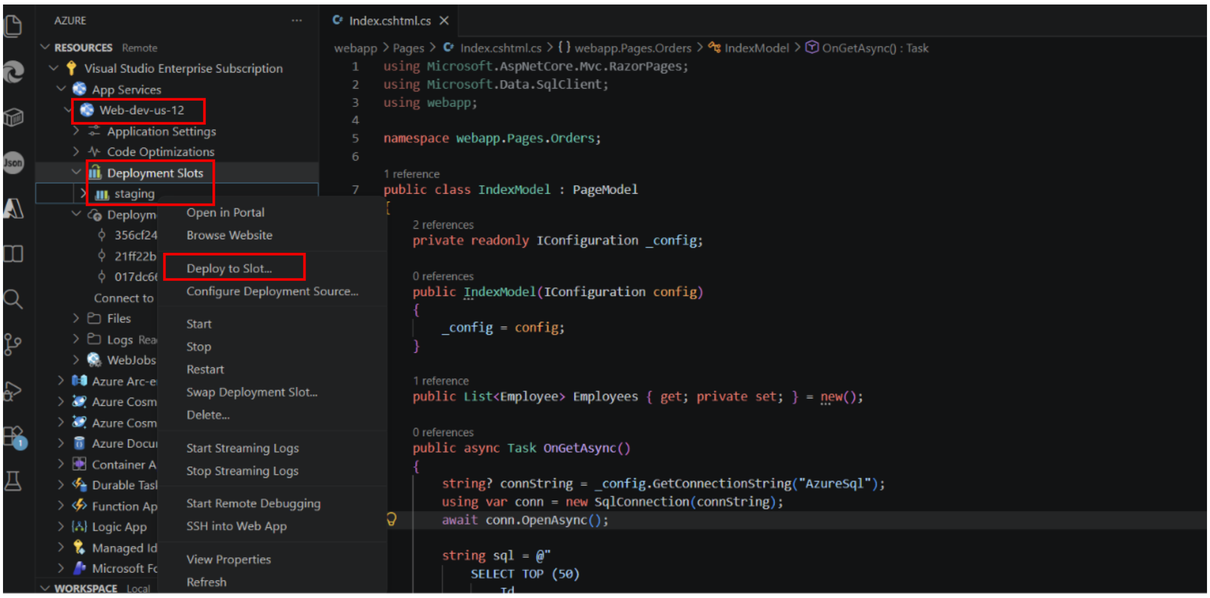
- Go to back **Azure Own WebApp.**
- Navigate to **Environment variables** **Connecting String.**
- Click on the **Add.**

- Enter the **connecting string name AzureSql.**
- Paste the **ADO.N ET (SQL authentication).**
- Enter the password in connecting string.
- Choose **SQLAzure.**
- Enable **Deployment slot setting.**
- Click on the **Apply**

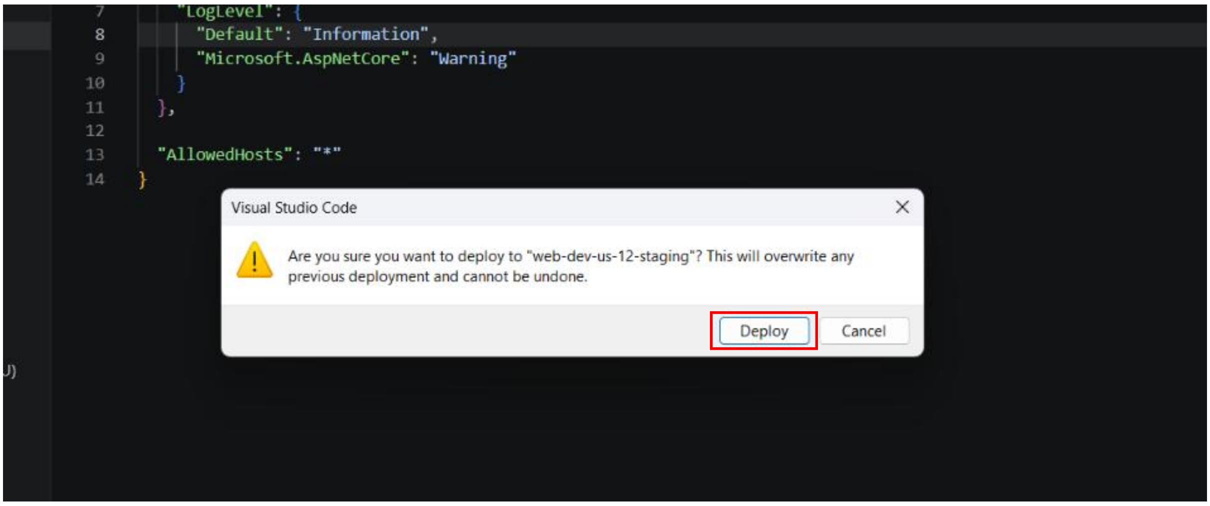


Step 6: Publish the Application to Staging

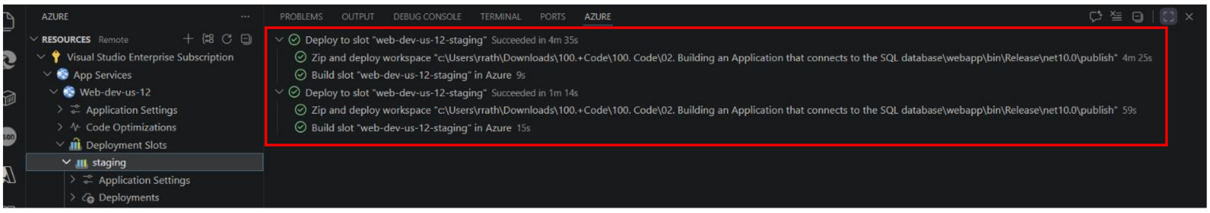
- Open the application in Visual Studio.
- Some changes in **Index.cshtml.cs** file in **code**
- Right-click the project and select **Publish**.
- Choose **Azure App Service**.
- Select the **Staging** deployment slot.
- Publish the application.



• click on the **Deploy.**



• Wait until **deployment completes** successfully.



Step 7: Verify the Staging Application

• Go to the **staging slot App service**.

• Navigate to **overview**.

• Copy the **Default Domain Url.**

• Paste the **Url in Browser**.

• Verify that the updated application is working correctly.

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Employee List

Id	Name	Email	Department	Salary
5	Neha Gupta	neha@example.com	Sales	55000.00
4	Amit Kumar	amit@example.com	Marketing	45000.00
3	Priya Singh	priya@example.com	Finance	60000.00
2	Rahul Sharma	rahul@example.com	HR	42000.00
1	Raman Rathaur	raman@example.com	IT	50000.00